

# Mitsue Project — Work Breakdown Structure

Phases 0–3 · April 2026 – September 2028

## 1. How to Read This Document

The WBS decomposes the entire project scope into discrete, manageable elements. Each element at the **lowest level** (the work package) represents a bounded unit of work with a defined deliverable, budget, and responsible party.

### Conventions used:

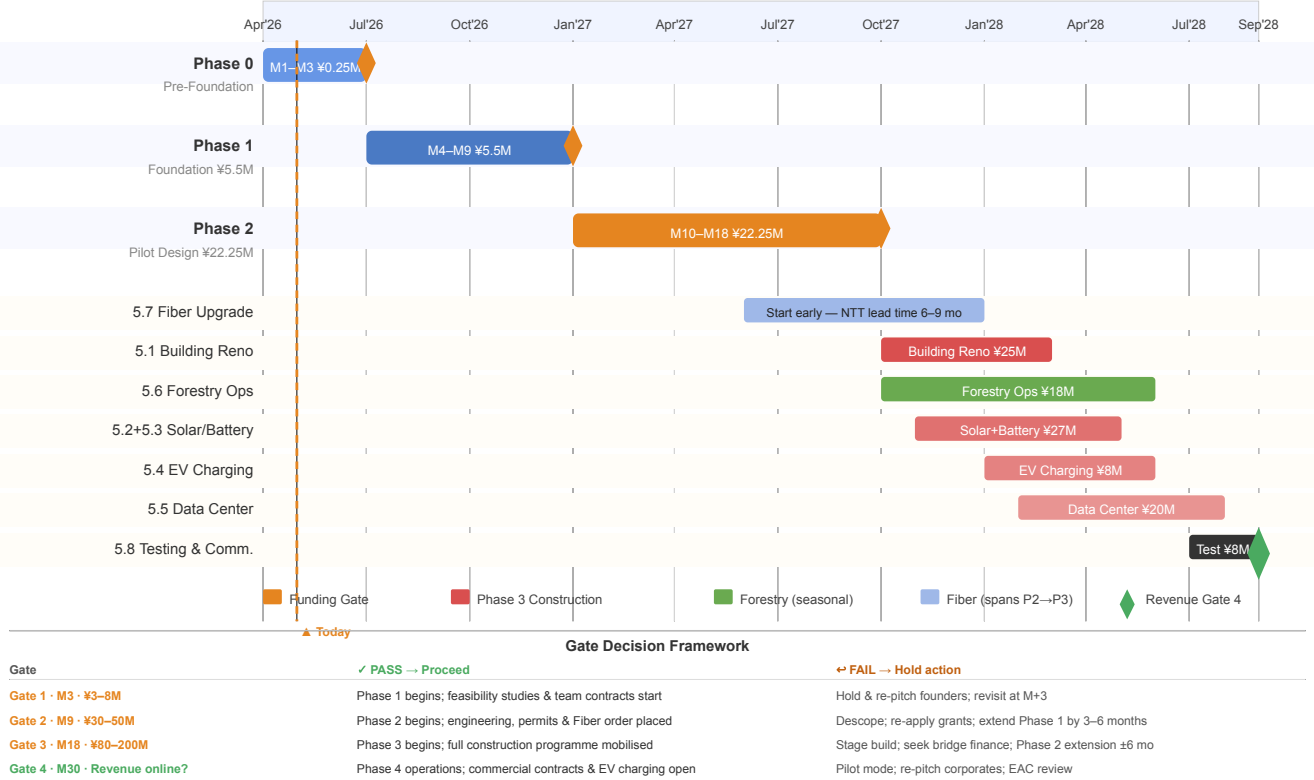
- WBS codes use dot notation: **1.0** = Level 1, **1.2** = Level 2, **1.2.3** = Level 3
- **Budget** columns show the planning estimate; ranges indicate remaining uncertainty
- **Owner** is the person accountable for delivery (not necessarily the doer)
- **Deliverable** is what must exist / be accepted for the element to be marked complete
- Total BAC = **¥220.0M** | Management Reserve (not in WBS) = ¥25.0M
- **Baseline Rev 1 (May 2026)** — Reality-checked against real-world Japan benchmarks for solar PV (¥200–300K/kW), school seismic retrofit (¥200–500K/m²), commercial EV fast-chargers (¥5–6M each), and forestry road works. BAC raised from ¥168M to ¥220M.

## 2. WBS Summary Tree

|     |                                       |         |
|-----|---------------------------------------|---------|
| 0.0 | MITSUE PROJECT (Phases 0-3)           | ¥220.0M |
| 1.0 | Project Management & Governance       | ¥15.0M  |
| 1.1 | Programme direction & reporting       |         |
| 1.2 | Legal, accounting & compliance        |         |
| 1.3 | Communications, translation & website |         |
| 2.0 | Phase 0 – Pre-Foundation              | ¥0.25M  |
| 2.1 | Community stakeholder engagement      |         |
| 2.2 | Founding charter & documents          |         |
| 2.3 | Founding team identification          |         |
| 3.0 | Phase 1 – Foundation                  | ¥5.5M   |
| 3.1 | Legal entity setup (一般社団法人)           |         |
| 3.2 | Forestry feasibility study            |         |
| 3.3 | Energy systems feasibility study      |         |
| 3.4 | Building & site assessment            |         |
| 3.5 | Connectivity assessment               |         |
| 3.6 | Operations setup                      |         |
| 4.0 | Phase 2 – Pilot Design                | ¥22.25M |
| 4.1 | Building & structural design          |         |
| 4.2 | Energy systems engineering            |         |
| 4.3 | Data center & IT architecture         |         |
| 4.4 | EV charging system design             |         |
| 4.5 | Permitting & regulatory               |         |
| 4.6 | Partnership & landowner agreements    |         |
| 4.7 | Grant writing & funding applications  |         |

|        |                       |                                  |         |
|--------|-----------------------|----------------------------------|---------|
|        | └─ 4.8                | Staff recruitment & onboarding   |         |
|        | └─ 4.9                | Vendor pre-qualification         |         |
|        | └─ 4.10               | Phase 2 contingency              |         |
|        |                       |                                  |         |
| └─ 5.0 | Phase 3 – Pilot Build |                                  | ¥177.0M |
|        | └─ 5.1                | School building renovation       |         |
|        | └─ 5.2                | Solar PV installation            |         |
|        | └─ 5.3                | Battery storage system           |         |
|        | └─ 5.4                | EV charging infrastructure       |         |
|        | └─ 5.5                | Data center fitout               |         |
|        | └─ 5.6                | Forestry operations              |         |
|        | └─ 5.7                | Fiber connectivity upgrade       |         |
|        | └─ 5.8                | Testing, commissioning & startup |         |
|        | └─ 5.9                | Phase 3 contingency              |         |

Figure 1 — Project Phase Timeline (Gantt View, Apr 2026 – Sep 2028)



### 3. WBS Dictionary — Full Detail

#### 1.0 Project Management & Governance — ¥15.0M

Cross-phase overhead covering leadership, legal compliance, and external communications. These costs are time-phased across all 30 months and do not correspond to a single phase.

##### 1.1 Programme Direction & Reporting — ¥8.0M

| Item  | Detail                                                                                                                     |
|-------|----------------------------------------------------------------------------------------------------------------------------|
| Scope | Representative Director oversight; monthly EVM reporting; board meetings; stakeholder briefings; risk register maintenance |
| Owner | Representative Director (代表理事)                                                                                             |

| Item         | Detail                                                                       |
|--------------|------------------------------------------------------------------------------|
| Budget       | ¥8.0M (¥265K/month average across 30 months)                                 |
| Deliverables | Monthly EVM status reports; gate review packages; annual stakeholder reports |
| Acceptance   | Board approval of each gate review package                                   |

**Work packages:**

- Monthly EVM status reports (M1–M30)
- Quarterly stakeholder briefings (8 reports)
- Four gate review packages (Gate 1–4)
- Annual year-end reviews (2026, 2027, 2028)
- Risk register (maintained monthly)
- OpenProject administration

**1.2 Legal, Accounting & Compliance — ¥4.0M**

| Item         | Detail                                                                                                         |
|--------------|----------------------------------------------------------------------------------------------------------------|
| Scope        | 行政書士 retainer for grants & permits; 税理士 for annual tax filing; legal counsel for contracts; ongoing compliance |
| Owner        | Director — Finance/Operations                                                                                  |
| Budget       | ¥4.0M                                                                                                          |
| Deliverables | Audited annual accounts; tax filings; legal review sign-offs on all major contracts                            |
| Acceptance   | Annual accounts accepted by auditor (監事)                                                                       |

**Work packages:**

- 行政書士 retainer (Phase 1–3)
- 税理士 / 公認会計士 annual engagement
- Annual financial report & audit
- Contract legal review (landowner, vendor, partnership agreements)

**1.3 Communications, Translation & Website — ¥3.0M**

| Item         | Detail                                                                                                              |
|--------------|---------------------------------------------------------------------------------------------------------------------|
| Scope        | Bilingual project website; press & stakeholder materials; translation of all formal documents; annual report design |
| Owner        | Director — Technology                                                                                               |
| Budget       | ¥3.0M                                                                                                               |
| Deliverables | Live bilingual website (by end of Phase 1); translated versions of all key documents; annual impact report          |
| Acceptance   | Website live and reviewed by board before Phase 1 close                                                             |

2.0 Phase 0 — Pre-Foundation — ¥0.25M

Duration: Month 1–3 (April–June 2026)

Low-budget, high-importance phase. The primary currency is time and trust, not money. Goal is to establish social license before any public commitments are made.

2.1 Community Stakeholder Engagement — ¥0.10M

| Item         | Detail                                                                                                                            |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Private meetings with village mayor, council members, 自治会 leaders in Sugano and nearby hamlets; listening sessions (not pitching) |
| Owner        | Rob Oudendijk + Japanese co-founder (when identified)                                                                             |
| Budget       | ¥0.10M (travel, hospitality, materials)                                                                                           |
| Deliverables | Meeting log with attendees and key reactions; summary memo of village leadership posture (supportive / neutral / cautious)        |
| Acceptance   | Village mayor informed; no active opposition identified                                                                           |

2.2 Founding Charter & Documents — ¥0.10M

| Item         | Detail                                                                                                             |
|--------------|--------------------------------------------------------------------------------------------------------------------|
| Scope        | Draft 2-page bilingual charter; preliminary project description in Japanese; translation of core concept documents |
| Owner        | Rob Oudendijk                                                                                                      |
| Budget       | ¥0.10M (translation fees, 行政書士 preliminary consultation)                                                           |
| Deliverables | Draft charter (日本語・英語); translated project concept note                                                            |
| Acceptance   | Charter reviewed and agreed by founding team members                                                               |

2.3 Founding Team Identification — ¥0.05M

| Item         | Detail                                                                                                                                          |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Identify and secure verbal commitments from 3–5 founding team members; at minimum one well-respected Japanese co-founder with rural credibility |
| Owner        | Rob Oudendijk                                                                                                                                   |
| Budget       | ¥0.05M (networking, travel)                                                                                                                     |
| Deliverables | Named list of founding team with verbal commitments documented                                                                                  |
| Acceptance   | Japanese Representative Director candidate identified                                                                                           |

**Phase 0 Gate Criterion:** Founding team agreed (verbal); village leadership informed and not opposed; draft charter complete.

3.0 Phase 1 — Foundation — ¥5.5M

Duration: Month 4–9 (July–December 2026)

The most structurally important phase. The feasibility studies produced here determine the credibility of all subsequent funding applications and Phase 2/3 budgets. Cutting corners here is false economy.

3.1 Legal Entity Setup — ¥0.5M

| Item         | Detail                                                                                                                                          |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | 一般社団法人 incorporation; notarized 定款; Legal Affairs Bureau (法務局) registration; opening bank account; accounting system setup (弥生会計 or equivalent) |
| Owner        | Director — Finance/Operations                                                                                                                   |
| Budget       | ¥0.5M (¥110K registration + ¥50K notary + ¥300K scrivener + ¥40K accounting setup)                                                              |
| Deliverables | Certificate of incorporation; registered office address; operational bank account; accounting system active                                     |

| Item       | Detail                                                      |
|------------|-------------------------------------------------------------|
| Acceptance | 法務局 registration confirmed; first board resolution recorded |

### 3.2 Forestry Feasibility Study — ¥1.5M

| Item         | Detail                                                                                                                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Field survey of candidate sugi plots for native species conversion; transportation and extraction cost analysis from steep terrain; native forest restoration plan, species selection, and timeline; carbon sequestration estimate; coordination with 日本林業協会 / 林野庁 advisors |
| Owner        | Director — Forestry/Local                                                                                                                                                                                                                                                   |
| Budget       | ¥1.5M (lower end of ¥1.5–3M range; full range covered by Phase 1 contingency if needed)                                                                                                                                                                                     |
| Deliverables | Forestry feasibility report (Japanese); executive summary (bilingual); land parcel map; carbon estimate methodology                                                                                                                                                         |
| Acceptance   | Report accepted by advisory board; J-Credit pre-qualification confirmed or ruled out                                                                                                                                                                                        |

### 3.3 Energy Systems Feasibility Study — ¥1.5M

| Item         | Detail                                                                                                                                                                                                                                                                                                                  |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Biomass CHP sizing (primary source) fuelled by sugi thinnings — baseload electricity + heat, fuel-supply logistics; complementary solar generation potential at school site; battery storage sizing; grid connection options; FIT/FIP eligibility assessment; EV charging load modelling; blackout resilience scenarios |
| Owner        | Director — Technology                                                                                                                                                                                                                                                                                                   |
| Budget       | ¥1.5M (lower end of ¥2–4M range)                                                                                                                                                                                                                                                                                        |
| Deliverables | Energy feasibility report; biomass CHP sizing and dispatch model; solar yield model; battery sizing recommendation; grid connection options memo; FIT/FIP eligibility letter                                                                                                                                            |
| Acceptance   | Report accepted by advisory board; METI pre-consultation completed                                                                                                                                                                                                                                                      |

### 3.4 Building & Site Assessment — ¥0.8M

| Item         | Detail                                                                                                                                                                                                     |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Structural condition survey of the Mitsue Taiken Koryukan and candidate factory building; seismic compliance assessment (新耐震基準); identification of required renovations for data center use; zoning review |
| Owner        | Director — Technology                                                                                                                                                                                      |
| Budget       | ¥0.8M (mid of ¥1–2M range for single-building survey)                                                                                                                                                      |
| Deliverables | Structural survey report; seismic compliance summary; renovation scope outline; zoning confirmation memo                                                                                                   |
| Acceptance   | Report accepted by board; Phase 3 renovation budget range confirmed                                                                                                                                        |

### 3.5 Connectivity Assessment — ¥0.4M

| Item         | Detail                                                                                                                         |
|--------------|--------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Current fiber capacity reaching Mitsue; upgrade requirements and costs; NTT consultation; satellite/microwave fallback options |
| Owner        | Director — Technology                                                                                                          |
| Budget       | ¥0.4M (mid of ¥0.3–0.5M range)                                                                                                 |
| Deliverables | Connectivity gap analysis; NTT meeting notes; upgrade cost estimate; backup options memo                                       |
| Acceptance   | Report accepted; Phase 3 fiber upgrade budget confirmed                                                                        |

### 3.6 Operations Setup — ¥0.8M

| Item  | Detail                                                                                                                                                                     |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope | Advisory board formalization (letters of commitment from Joi Ito, Ray Ozzie, others); initial bilingual website live; funding pipeline documentation; grant calendar setup |

| Item         | Detail                                                                              |
|--------------|-------------------------------------------------------------------------------------|
| Owner        | Representative Director                                                             |
| Budget       | ¥0.8M                                                                               |
| Deliverables | Signed advisory board letters; live website; grant application calendar for Phase 2 |
| Acceptance   | Website reviewed by board; advisory board confirmed in writing                      |

**Phase 1 Gate Criterion:** Legal entity registered; all four feasibility studies accepted; ¥3–8M secured; letter of support from Mitsue village government.

4.0 Phase 2 — Pilot Design — ¥22.25M  
*Duration: Month 10–18 (January–September 2027)*

Converts feasibility into engineering reality. The permit applications submitted here are the long-lead items that drive Phase 3 start. Staffing up in this phase is critical.

**4.1 Building & Structural Design — ¥2.0M**

| Item         | Detail                                                                                                                                                                                |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Full architectural drawings for school renovation (1 wing); seismic retrofit specifications; mechanical/electrical/plumbing for data center wing; building permit application package |
| Owner        | Director — Technology                                                                                                                                                                 |
| Budget       | ¥2.0M                                                                                                                                                                                 |
| Deliverables | Stamped architectural drawings; MEP specs; building permit application submitted                                                                                                      |
| Acceptance   | Building permit application accepted by 奈良県 authorities                                                                                                                               |

**4.2 Energy Systems Engineering — ¥2.0M**

| Item         | Detail                                                                                                                                                               |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Detailed design for solar PV array, battery storage, EV charging integration, and grid connection; equipment specifications; load flow analysis; FIT/FIP application |
| Owner        | Director — Technology                                                                                                                                                |
| Budget       | ¥2.0M                                                                                                                                                                |
| Deliverables | Detailed energy system design package; equipment spec sheets; FIT/FIP application submitted to METI                                                                  |
| Acceptance   | METI FIT/FIP application reference number received                                                                                                                   |

**4.3 Data Center & IT Architecture — ¥1.0M**

| Item         | Detail                                                                                                                              |
|--------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Server room layout; power and cooling design for 10–20 servers; network architecture; APPI compliance plan; cybersecurity framework |
| Owner        | Director — Technology                                                                                                               |
| Budget       | ¥1.0M                                                                                                                               |
| Deliverables | Data center design package; network topology diagram; security framework document                                                   |
| Acceptance   | Design reviewed and accepted by technical advisory committee                                                                        |

**4.4 EV Charging System Design — ¥1.0M**

| Item         | Detail                                                                                                                |
|--------------|-----------------------------------------------------------------------------------------------------------------------|
| Scope        | Charging station layout for 4 units; grid integration design; 消防 and 電気設備 permit preparation; signage and access plan |
| Owner        | Director — Technology                                                                                                 |
| Budget       | ¥1.0M                                                                                                                 |
| Deliverables | EV charging design drawings; permit pre-application submitted                                                         |
| Acceptance   | Fire department and electrical safety pre-consultation completed                                                      |

**4.5 Permitting & Regulatory — ¥3.5M**

| Item  | Detail                                                                                                                                                                                                            |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope | Manage all permit applications across agencies: building permit (建築基準法), forestry cutting notification (伐採届出), FIT/FIP registration (METI), EV charging safety permits (消防・電気), environmental review coordination |
| Owner | Director — Finance/Operations (supported by 行政書士)                                                                                                                                                                 |

| Item         | Detail                                                                                    |
|--------------|-------------------------------------------------------------------------------------------|
| Budget       | ¥3.5M (includes 行政書士 fees of ¥200–500K per application)                                   |
| Deliverables | All permit applications submitted; building permit issued; FIT/FIP registration confirmed |
| Acceptance   | Core permits in hand before Phase 3 procurement begins                                    |

**Note:** METI FIT/FIP registration and building permits are the critical-path items — typical lead time 4–6 months. Applications must be submitted no later than M12 (March 2027).

#### 4.6 Partnership & Landowner Agreements — ¥1.5M

| Item         | Detail                                                                                                                                                                                   |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Negotiate and execute landowner contracts for sugi harvesting rights; memoranda of understanding with key technical and institutional partners; village government cooperation agreement |
| Owner        | Representative Director                                                                                                                                                                  |
| Budget       | ¥1.5M (legal drafting, negotiation facilitation, translation)                                                                                                                            |
| Deliverables | Minimum 2 signed landowner harvest agreements; village government MOU signed; at least 1 corporate partner MOU                                                                           |
| Acceptance   | Legal review sign-off on all agreements; board approval                                                                                                                                  |

#### 4.7 Grant Writing & Funding Applications — ¥2.0M

| Item         | Detail                                                                                                                                                                   |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Prepare and submit applications for major grants: 地方創生関係交付金, 林野庁 subsidies, NEDO, METI green tech, Nara Prefecture, Nippon Foundation, Japan Fund for Global Environment |
| Owner        | Director — Finance/Operations (supported by specialist 行政書士)                                                                                                             |
| Budget       | ¥2.0M (行政書士 fees ¥200–500K each; 4–6 applications)                                                                                                                       |
| Deliverables | Minimum 4 grant applications submitted; at least ¥30M in pending applications by end of Phase 2                                                                          |
| Acceptance   | Applications submitted; Gate 3 funding of ¥80–200M confirmed or committed                                                                                                |

#### 4.8 Staff Recruitment & Onboarding — ¥6.0M

| Item         | Detail                                                                                                                                        |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Hire first 2–3 part-time staff (project coordinator, operations/admin, possibly local forestry liaison); onboarding and process documentation |
| Owner        | Representative Director                                                                                                                       |
| Budget       | ¥6.0M (9 months × 3 × ~¥220K/month blended part-time)                                                                                         |
| Deliverables | 2–3 staff under contract; onboarding complete; key processes documented                                                                       |
| Acceptance   | Staff in post and contributing before Phase 3 begins                                                                                          |

#### 4.9 Vendor Pre-Qualification — ¥1.5M

| Item         | Detail                                                                                                                                                                                           |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Shortlist and pre-qualify vendors for: building renovation contractor, solar/battery supplier, data center hardware, EV charging equipment; obtain binding budget quotes for Phase 3 procurement |
| Owner        | Director — Technology                                                                                                                                                                            |
| Budget       | ¥1.5M                                                                                                                                                                                            |
| Deliverables | Vendor shortlist per category; budget quotes received; Phase 3 procurement plan                                                                                                                  |
| Acceptance   | Phase 3 WBS 5.x budgets confirmed within ±15% of EVM baseline before Gate 3                                                                                                                      |

#### 4.10 Phase 2 Contingency — ¥1.75M

| Item | Detail |
|------|--------|
|------|--------|



| Item   | Detail                                                                                                                                                            |
|--------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope  | Reserve for Phase 2 cost variances — particularly permit fees higher than estimated, additional engineering iterations, or grant application costs exceeding plan |
| Owner  | Representative Director                                                                                                                                           |
| Budget | ¥1.75M (~8% of Phase 2 direct costs)                                                                                                                              |
| Access | Requires Representative Director approval; drawn down only against specific identified variances                                                                  |

**Phase 2 Gate Criterion:** ¥30–50M secured; detailed engineering complete; key permits in hand (or application submitted with favourable pre-consultation); 2 staff in post.

5.0 Phase 3 — Pilot Build — ¥177.0M

Duration: Month 19–30 (October 2027 – September 2028)

The largest and most complex phase. Peak spend is in Months 22–27. All Phase 3 work packages are interdependent — building renovation must precede data center fitout; solar/battery must precede EV commissioning.

Sequencing logic:

5.1 Building Renovation → 5.5 Data Center Fitout

5.2 Solar PV + 5.3 Battery → 5.4 EV Charging commissioning

5.6 Forestry Operations → independent track (weather/season dependent)

5.7 Fiber → early in P3 (long lead time with NTT)

5.8 Testing & Commissioning → final 2 months

5.1 School Building Renovation — ¥38.0M

| Item         | Detail                                                                                                                                                                                                                 |
|--------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Seismic retrofit of 1 wing; internal renovation for data center and office use; MEP installation (power distribution, cooling, fire suppression); accessible entrance and welfare facilities; exterior weatherproofing |
| Owner        | Director — Technology                                                                                                                                                                                                  |
| Budget       | ¥38.0M                                                                                                                                                                                                                 |
| Deliverables | Renovated wing with 建築基準法 compliance certificate; handover from contractor; punch-list sign-off                                                                                                                        |
| Acceptance   | Building inspector sign-off; internal acceptance by project board                                                                                                                                                      |
| Key risks    | Unforeseen structural issues adding scope; labour shortage in rural Nara                                                                                                                                               |

Work packages:

- Seismic retrofit (structural steel/concrete)
- MEP rough-in (power, cooling, fire suppression)
- Internal partitioning and fit-out
- Exterior repairs and weatherproofing
- Site works (access road, parking, landscaping for EV)
- Final inspections and compliance certificates

5.2 Solar PV Installation — ¥22.0M

| Item         | Detail                                                                                                                                     |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Rooftop solar PV (~100 kW); inverters; monitoring system; grid connection infrastructure; commissioning and FIT/FIP handover documentation |
| Owner        | Director — Technology                                                                                                                      |
| Budget       | ¥22.0M (approx. ¥220K/kW installed, rural premium)                                                                                         |
| Deliverables | Grid-connected solar array generating power; FIT/FIP connection agreement in place; monitoring dashboard live                              |
| Acceptance   | METI FIT/FIP connection confirmed; generation data logging active                                                                          |

5.3 Battery Storage System — ¥12.0M (subject to feasibility study confirmation)

| Item  | Detail                                                                                                                                                                                                                                                                               |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope | Lithium-ion battery storage system sized for 12–48 hours critical-facility backup; BMS; integration with solar and grid; safety certifications. <b>Decision gate: Phase 1 energy feasibility study must confirm economic and operational case before this work package proceeds.</b> |

| Item         | Detail                                                                                        |
|--------------|-----------------------------------------------------------------------------------------------|
| Owner        | Director — Technology                                                                         |
| Budget       | ¥12.0M                                                                                        |
| Deliverables | Battery system installed, commissioned, and safety-certified; blackout simulation test passed |
| Acceptance   | 消防 safety certificate; 48-hour backup test at design load                                     |

#### 5.4 EV Charging Infrastructure — ¥15.0M

| Item         | Detail                                                                                                                                                          |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | 4 EV charging stations (mix of standard AC and fast DC); civil works for cable trenching; payment/management software; signage; safety approvals; public launch |
| Owner        | Director — Technology                                                                                                                                           |
| Budget       | ¥15.0M                                                                                                                                                          |
| Deliverables | 4 operational charging stations open to public; payment system live; 消防・電気設備 approvals in hand                                                                  |
| Acceptance   | Public launch event; village government acceptance                                                                                                              |

#### 5.5 Data Center Fitout — ¥20.0M

| Item         | Detail                                                                                                                                                                                                                     |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Server racks and hardware (10–20 servers, edge computing focus); precision cooling; structured cabling; UPS backup integration with battery system; network equipment; APPI-compliant security controls; remote monitoring |
| Owner        | Director — Technology                                                                                                                                                                                                      |
| Budget       | ¥20.0M                                                                                                                                                                                                                     |
| Deliverables | Operational data center accepting first paying workloads; monitoring dashboard live; APPI compliance documentation complete                                                                                                |
| Acceptance   | First commercial hosting contract signed; uptime monitoring active                                                                                                                                                         |

#### 5.6 Forestry Operations — ¥25.0M

| Item         | Detail                                                                                                                                                                                                       |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | First 5–10 ha of sugi harvested under signed landowner agreements; timber processed for lumber; native species replanting commenced; forest road and drainage improvements; J-Credit documentation initiated |
| Owner        | Director — Forestry/Local                                                                                                                                                                                    |
| Budget       | ¥25.0M                                                                                                                                                                                                       |
| Deliverables | Harvest completion report; replanting plan executed for cleared area; first J-Credit application submitted                                                                                                   |
| Acceptance   | 伐採届出 (cutting notification) filed and acknowledged; replanting confirmed by site inspection                                                                                                                  |
| Key risks    | Seasonal access (avoid typhoon season M19–M23); steep terrain requiring specialised equipment                                                                                                                |

#### 5.7 Fiber Connectivity Upgrade — ¥10.0M

| Item         | Detail                                                                                                                                                                                 |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Upgrade fiber capacity to Mitsue in coordination with NTT and/or regional ISP; install dedicated dark fiber or upgraded leased line to school site; microwave backup link installation |
| Owner        | Director — Technology                                                                                                                                                                  |
| Budget       | ¥10.0M                                                                                                                                                                                 |
| Deliverables | Confirmed bandwidth upgrade (target: 1 Gbps symmetric minimum to school site); backup link live; NTT SLA in place                                                                      |
| Acceptance   | Bandwidth test confirming contracted speeds; failover test on backup link                                                                                                              |

| Item | Detail                                                                         |
|------|--------------------------------------------------------------------------------|
| Note | NTT coordination has 6–9 month lead time — initiate in Phase 2 (M15 at latest) |

### 5.8 Testing, Commissioning & Startup — ¥8.0M

| Item         | Detail                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope        | Integrated system testing across all Phase 3 elements; punch-list resolution; staff training; documentation finalisation; public launch event; Phase 4 operational handover |
| Owner        | Representative Director                                                                                                                                                     |
| Budget       | ¥8.0M                                                                                                                                                                       |
| Deliverables | All systems tested and operational; staff trained; punch-list closed; Phase 4 operational plan approved; launch event completed                                             |
| Acceptance   | Board acceptance of Phase 3 completion; Phase 4 operational plan approved                                                                                                   |

#### Sub-work packages:

- Solar + battery integrated test (blackout simulation)
- Data center stress test and security audit
- EV charging operational test (all 4 stations)
- Forestry first-year progress review
- Fiber latency and failover test
- Staff training programme (data center ops, EV maintenance, forestry coordination)
- Phase 3 completion documentation package
- Public launch event (village + media + funders)

### 5.9 Phase 3 Contingency — ¥27.0M

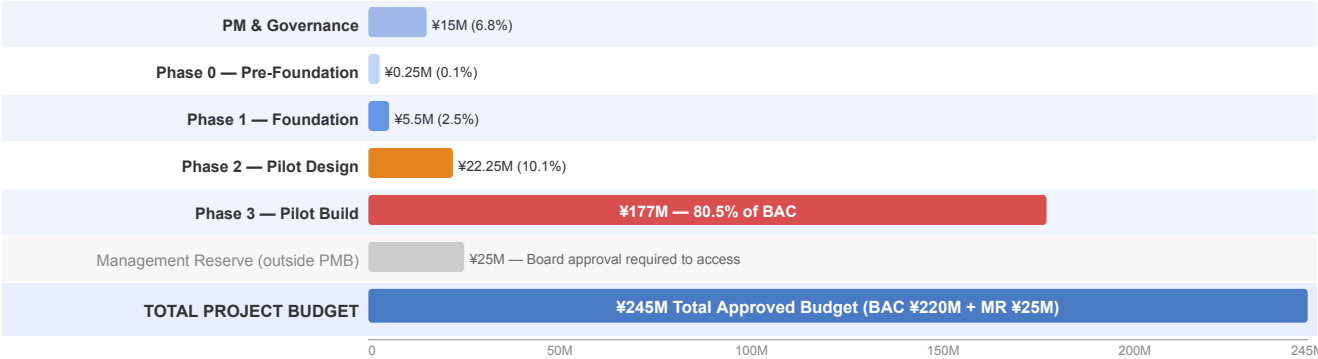
| Item   | Detail                                                                                                                                         |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope  | Reserve for Phase 3 cost variances — primarily structural findings in 5.1, equipment cost movements, rural labour premiums, or seasonal delays |
| Owner  | Representative Director                                                                                                                        |
| Budget | ¥27.0M (~18% of Phase 3 direct costs)                                                                                                          |
| Access | Representative Director approval for draws up to ¥5M; board approval required above ¥5M                                                        |

**Phase 3 Gate Criterion (Gate 4):** All systems operational; first commercial data center contract signed; EV charging open to public; forestry harvest commenced; Phase 4 plan approved by board.

## 4. Budget Summary

| WBS | Element                         | Budget (¥M)   | % of BAC    |
|-----|---------------------------------|---------------|-------------|
| 1.0 | Project Management & Governance | 15.00         | 6.8%        |
| 2.0 | Phase 0 — Pre-Foundation        | 0.25          | 0.1%        |
| 3.0 | Phase 1 — Foundation            | 5.50          | 2.5%        |
| 4.0 | Phase 2 — Pilot Design          | 22.25         | 10.1%       |
| 5.0 | Phase 3 — Pilot Build           | 177.00        | 80.5%       |
|     | <b>Total BAC (PMB)</b>          | <b>220.00</b> | <b>100%</b> |
|     | Management Reserve (not in PMB) | 25.00         | —           |
|     | <b>Total Project Budget</b>     | <b>245.00</b> | —           |

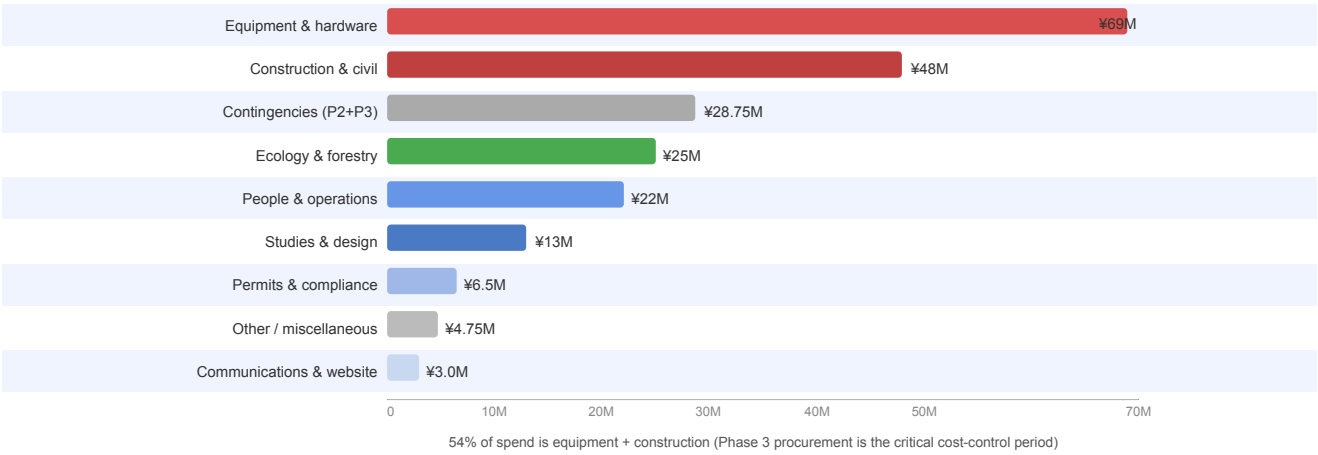
Figure 2 — Budget by WBS Element



Budget by Category (across all phases)

| Category              | Budget (¥M) | Examples                                                 |
|-----------------------|-------------|----------------------------------------------------------|
| Studies & design      | ¥13.0M      | Feasibility studies, engineering design, IT architecture |
| Construction & civil  | ¥48.0M      | Building renovation, site works, civil for EV/fiber      |
| Equipment & hardware  | ¥69.0M      | Solar, battery, EV chargers, servers, network gear       |
| Ecology & forestry    | ¥25.0M      | Harvest operations, replanting, forest roads             |
| Permits & compliance  | ¥6.5M       | 行政書士 fees, permit applications, legal                    |
| People & operations   | ¥22.0M      | PM team, hired staff, advisory board                     |
| Contingencies         | ¥28.75M     | Phase 2 and Phase 3 contingency reserves                 |
| Communications        | ¥3.0M       | Website, translation, design                             |
| Other / miscellaneous | ¥4.75M      | Accounting, banking, training, launch event              |
| Total BAC             | ¥220.0M     |                                                          |

Figure 3 — Budget Breakdown by Cost Category



5. Key Dependencies Map

| Dependency               | Predecessor | Successor              | Consequence if late                       |
|--------------------------|-------------|------------------------|-------------------------------------------|
| Japanese co-founder      | 2.3         | 3.1, 3.6               | Cannot incorporate without Rep. Director  |
| Village govt support     | 2.1         | 3.1, 4.6               | No MOU; weakens grant applications        |
| Feasibility studies      | 3.2–3.5     | 4.1–4.4, gate funding  | Phase 2 budgets unreliable; grants harder |
| METI FIT/FIP application | 4.2         | 5.2 commissioning      | 6-month METI review — must submit by M12  |
| Building permit          | 4.1         | 5.1 construction start | Cannot commence construction without it   |
| NTT fiber initiation     | 4.5 / 4.7   | 5.7                    | 6–9 month NTT lead time                   |

| Dependency          | Predecessor | Successor | Consequence if late                                |
|---------------------|-------------|-----------|----------------------------------------------------|
| Gate 3 funding      | 4.7         | 5.0 (all) | Phase 3 cannot start without confirmed ¥80M+       |
| Building renovation | 5.1         | 5.5       | Data center cannot be fitted before shell complete |
| Solar + battery     | 5.2 + 5.3   | 5.4       | EV charging dependent on local power supply        |

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